Power

antocciphysics | TEACHER KEY | CPA

Learning target: Compare how quickly energy is transferred.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 13; 3 course days later.*
Difference 6 J may be rotational energy and thermal energy; other justified transfers acceptable.

02. Explain a worked example (worked)
Energy equal; shorter time means greater power.

03. Look and explain (model)
First; 60 W versus 20 W.

04. Build the solution (practice)
30 W; 180 J.

05. Use the idea (reason)
100 J per second; energy=power×specified duration.

06. Choose a method (mixed) *Review: Unit 4, Day 4; 12 course days later.*
W_{net}=-20 J; speed falls for fixed mass.

07. Exit check (exit)
Same work; faster student twice average power.

08. STRETCH: Apply it another way (stretch)
Examples 100 J in 2 s and 250 J in 5 s.

Power

antocciphysics | TEACHER KEY | HONORS

Learning target: Compare how quickly energy is transferred.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 13; 3 course days later.*
mgh≈1.47 J; K=0.3375 J; fraction≈0.2296 (22.96%).

02. Explain a worked example (worked)
Energy equal; shorter time means greater power.

03. Represent and interpret (model)
First; 60 W versus 20 W.

04. Solve and justify (practice)
2352 J; 294 W.

05. Analyze the reasoning (reason)
100 J per second; energy=power×specified duration.

06. Choose a method (mixed) *Review: Unit 4, Day 4; 12 course days later.*
$W_{net}=½m(v_f^2-v_i^2)=-56$ J, negative as kinetic energy falls.

07. Exit check (exit)
600 W; 3000 J.

08. STRETCH: Apply it another way (stretch)
Useful 150 W; input 200 W; input 800 J, useful 600 J, other 200 J.